TARO M. SHEFFERSON-NAGATA

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EDUCATION

Princeton University Expected May 2030
B.S.E. candidate; coursework: Multivariable Calculus, Computer Science, Physics
The University of Tokyo: Natural Sciences I Apr–Aug 2026
Meikei Gakuen High School: IB Diploma 45/45; SAT 1600 2026

HONORS AND SCHOLARSHIPS

- **Japan Math A-Lympiad**: **3rd nationally**.
- **Japan AI Education Utilization Award**: Grand Prize; **1st nationally** among 477 teams.
- **National Science and Research Paper Award**: Outstanding Prize; **2nd nationally** among 160+ teams.
- **Yanai Tadashi Foundation Scholarship (2025)**: Four-year, full-expense; approximately **US$460,000**.
- **Event Koshien (2025)**: Finalist; **5th in Japan**.
- **GEC Business Competition**: **Top 8 nationally** in Japan.

TECHNICAL SKILLS

Programming: Python, R, Fortran, JavaScript, Java,
Web: HTML, CSS | **Hardware:** Raspberry Pi
Scientific computing: JAX, DiffHydro, numerical modeling, simulation, digital twins
Languages: Japanese (native); English (fluent; IELTS Academic 7.5)

RESEARCH AND QUANTITATIVE EXPERIENCE

SpaceData Inc. Sep 2026–Present
AI and Digital Twin Development
Company technology: PROVIDENCE; PLATEAU city data, OpenUSD / NVIDIA Omniverse, and Python / Houdini automation.

- Building digital twins of Tokyo for computational modeling and analysis of urban operations.
- Developing and applying SpaceData AI to analyze city models and guide operational improvements.

National Astronomical Observatory of Japan (NAOJ) 2025–Present
Research Collaborator with Kazumasa Ohno

- Developed a radiative-equilibrium model of cloud-particle temperatures across eight condensate species, varying stellar spectra, irradiation levels, and particle sizes.
- Mapped composition-dependent stability regimes, finding that silicate clouds remain stable over broad conditions while sulfide clouds can overheat and sublimate.
- Applied the model to WASP-17b and HD 189733b, obtaining results consistent with JWST observations and predictions testable with future observations.
- Led the analysis, interpretation, writing, and peer-review revisions for a first-author paper in *The Astrophysical Journal*.

Kavli IPMU, The University of Tokyo Aug 2026–Present
Research Intern with Enrico Garaldi

- Conducting computational astrophysics and machine-learning research using **DiffHydro**, a **Python/JAX**-based differentiable hydrodynamics framework.
- Investigating gradient-based parameter optimization and inverse problems through differentiable physical simulation.
- Developing code, numerical tests, and example simulations to improve and validate the radiative-transfer module.

PEER-REVIEWED PUBLICATION

Shefferson-Nagata, T. M., and Ohno, K. (2026). "Radiatively Controlled Thermal Stability of High-Altitude Clouds in Exoplanetary Atmospheres." *The Astrophysical Journal*, 1005, 210. doi:10.3847/1538-4357/ae7d27.

Presentations: JpGU–AGU Joint Meeting 2026 (oral); Japan Geoscience Union Meeting, University of Tokyo, 2025 (poster).

AI AND ENTREPRENEURIAL PROJECTS

NOBILVA

Founder and AI Product Developer

- Developed an AI learning chatbot for children facing barriers to attending school, supporting learners across Japan; won the Japan AI Education Utilization Award's national Grand Prize (1st among 477 teams).
- Received Japanese government grant funding and institutional backing, including promotional support.
- Founded and scaled the organization to 17 team members, supporting more than 150 students.

Thy & I 2026

Co-founder and CTO

- Built the website, booking, and payment systems for a university-admissions service.
- Designed a university-admissions program of three individualized 90-minute sessions.

PRINCETON ACTIVITIES

Quantitative Traders (PQT): Three-stage selection, including a mathematics test ($< 10\%$ acceptance).

Student Ventures (PSV): One of three new members; three-stage selection ($< 5\%$ acceptance).

Robotics Club: Building a Star Wars BB-8 droid with the club.